

When Can a Machine Judge Like a Human?

Evaluating LLM-as-a-Judge Approaches for Human-Centered Computing Research

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Abstract

Large language models (LLMs) are flooding into Human-Centered Computing (HCC) research as substitutes for human judgment: coding qualitative data, annotating crowdsourced corpora, simulating study participants, generating personas, and evaluating designs. Yet HCC’s methods carry epistemological commitments (interpretation, reflexivity, positionality) that make uncritical substitution risky, and the field lacks an evidence-based account of *when* an “LLM-as-a-judge” can stand in for a human and when it cannot. We address this with four studies. **Study 1a** is a PRISMA systematic review of 217 studies across 29 domains that establishes the general reliability landscape of LLM-as-a-judge: substitution is validated ($\approx$ or above the human-human ceiling) for objective, verifiable, low-expertise tasks judged in aggregate, and unreliable for subjective, expert, low-resource, or adversarial ones (mean reliability 3.15/5; only 31% of studies in the reliable band). **Study 1b** asks whether this landscape is a moving target: we re-run a flagship “promising” benchmark (SummEval) with the most powerful open models served via Clemson’s vLLM API and ask whether they now *surpass* the agreement numbers reported for proprietary judges, distinguishing a “matter of time” from a structural ceiling. **Study 2** is a PRISMA systematic review of 55 studies focused on HCC, HCI, and qualitative-research methods that historically require human input. It finds that the field overwhelmingly positions LLMs to *augment* rather than replace humans (36 augment vs. 14 replace vs. 5 simulate), with an epistemic stance dominated by *tension* (38/55); objective crowd-annotation substitution is validated, interpretive qualitative analysis is reliable only as human-in-the-loop augmentation, and *simulating* participants, personas, and usability evaluations is the least reliable (flattening of identity groups, caricature, hallucinated usability issues). **Study 3** closes the gaps the first two studies expose, by running a standardized LLM-as-a-judge meta-evaluation, using open models served via Clemson’s vLLM API and the agreement-against-human-ceiling methodology distilled from Studies 1-2, on HCC datasets that contain human labels but have never been formally tested with an LLM judge. Across 18000 judgments (a 6-model open panel, 9B-754B), it finds the human-agreement ceiling governs reliability: the panel matches or exceeds humans on low-ceiling NLI but falls short of, and fails the alt-test for, subjective social-computing annotation (emotion, hate speech), with binary offensiveness the strongest case; a recent low-contamination irony set is matched only at its own low human ceiling, ruling out memorization as the explanation. A mixed-effects model confirms the mechanism: judge correctness is governed overwhelmingly by item-level human (dis)agreement (random-intercept $SD_{\text{item}} = 2.02$) and by task subjectivity ($OR = 0.17$ per unit of 1–ceiling), while model scale contributes only modestly ($OR = 1.30$ per $\log_{10}$ params); after FDR correction 22 of 36 judge $\times$ dataset cells sit significantly below the human ceiling. Together the studies yield a decision framework telling HCC researchers when an LLM judge is a validated substitute, a human-in-the-loop assistant, or an unsafe replacement.

*Correspondence: ai4helab@gmail.com. This research used an AI-assisted systematic-search and extraction pipeline; all search logs (OpenAlex, Semantic Scholar, web), the two extracted study corpora, the analysis code, the figure/table-generation scripts, and the Study-3 evaluation harness are released for full reproducibility.

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1 Introduction and motivation

In barely two years, LLMs have moved from objects of HCI study to *instruments* of HCI method. Researchers now use them to code interview transcripts [17, 46], run thematic analyses [12, 37], annotate social-media corpora [18, 41], simulate survey respondents and “silicon samples” [2], generate personas [38], and even perform heuristic usability evaluation [20]. The appeal is obvious: human judgment is the costly bottleneck of human-centered research, and an LLM that labels for \$0.003 an item is a tempting stand-in [18].

Study 1a: <i>Where</i> is LLM-as-a-judge reliable? (cross-domain review, 217 studies) → Study 1b: <i>Is it a moving target?</i> (most powerful open models vs reported numbers, SummEval) → Study 2: <i>How does HCC use LLMs</i> for human-input tasks? (55 studies) → Study 3: <i>Fill the gaps</i>, standardized evaluation on HCC datasets with human labels (open models via Clemson vLLM).

Figure 1: The four-study progression. Study 1a maps where LLM judges are reliable; Study 1b tests whether today’s strongest open models move that map; Study 2 narrows to Human-Centered Computing practice; Study 3 supplies new empirical evidence on under-tested HCC datasets.

The danger is equally clear. HCC and qualitative methods are not mere labeling pipelines; they carry *epistemological commitments* (interpretation, reflexivity, positionality, and the situated knowledge of the researcher) that an LLM cannot obviously honor [13, 40]. Substituting a model for a human participant can *flatten* and misportray the very identity groups HCI seeks to center [10, 44]. And the empirical record is genuinely mixed: the same model can match two human coders on a structured codebook [15] yet hallucinate usability problems that do not exist [20], or agree with a second annotator on stance [18] yet be fooled into mislabeling relevance by a few injected keywords [1].

HCC researchers therefore face a question with no consolidated answer: **for which HCC research tasks is an LLM-as-a-judge a safe substitute for human judgment, for which is it a useful assistant, and for which is it unsafe?** This paper answers that question through three studies with a deliberate progression (Figure 1):

1. **Study 1a, the general landscape.** A PRISMA systematic review of 217 studies across 29 domains establishes *where, in general*, LLM-as-a-judge agrees with humans, anchored to the human-human ceiling. This gives HCC the broad map and the methodological vocabulary (the reliability rubric, the bias catalogue, the decision map).
2. **Study 1b, is the landscape a moving target?** The review captures results *as reported*, often with one- or two-generation-old proprietary judges. We re-run a flagship “promising”-band benchmark (SummEval summarization quality) with the *most powerful open models available today* and ask whether they now surpass the canonical reported judge correlations, separating limitations that are a “matter of time” (scale/recency) from *structural* ceilings that more capable models do not cross.
3. **Study 2, the HCC lens.** A second PRISMA review of 55 studies zooms into the methods HCC actually uses (qualitative coding, thematic/content analysis, crowdsourced annotation, participant simulation, persona generation, usability evaluation, and design critique) and codes how the field positions LLMs (augment / replace / simulate) and with what epistemic stance.
4. **Study 3, filling the gaps.** Studies 1-2 reveal HCC datasets that contain human labels but have never been formally evaluated with an LLM judge. Study 3 runs a standardized LLM-as-a-judge meta-evaluation on these datasets, using open models served through Clemson’s vLLM API and the agreement-against-human-ceiling methodology distilled from the prior literature, with a mixed-effects analysis of what governs judge reliability.

Contributions. (1) The first consolidated, human-ceiling-anchored map of LLM-as-a-judge reliability across 29 domains (Study 1a). (2) A direct test of whether the most powerful open models available today surpass the reported judge correlations in a flagship “promising” domain, adjudicating “matter of time” vs. structural ceiling (Study 1b). (3) The first systematic review of LLM substitution

specifically for HCC/HCI/qualitative methods, with a method×role×stance coding that surfaces the field’s augment-not-replace consensus and its epistemic tensions (Study 2). (4) A reproducible, standardized evaluation harness plus a mixed-effects account of what governs judge reliability (Study 3), tested on HCC datasets that had human labels but no prior LLM-judge evaluation, plus two released corpora (217 + 55 coded studies), all code, and a decision framework for practitioners.

2 Background and shared methodology

LLM-as-a-judge descends from reference-free metrics (BERTScore, COMET) but uses a general instruction-following model with a natural-language rubric, supporting pointwise, pairwise, and listwise judgment [19, 28]. Four architectural families recur: prompted general LLMs [31, 48]; fine-tuned open evaluators [26, 43, 50]; juries/multi-agent panels [9, 42]; and calibrated rubric methods [21]. Prior surveys taxonomize methodology [19, 29]; we instead synthesize *where substitution is empirically validated*, and, uniquely, do so through the HCC lens.

Shared protocol. Both reviews follow PRISMA 2020 [35], identify studies through a triangulated search (the OpenAlex API as the primary bibliographic database, Semantic Scholar as a cross-check, and a structured web “deep-research” arm), and code each study on a five-level **reliability verdict** benchmarked against the *human-human agreement ceiling* for the same task: **1 Validated** ($\approx$ / $>$ ceiling), **2 Promising** (conditional), **3 Mixed** (task-dependent), **4 Caution** (below ceiling), **5 Unreliable** (near chance / bias-dominated). We report a reliability score = 6 − verdict and synthesize within metric families rather than pooling incommensurable effect sizes. Full protocols, search logs, and the two coded corpora are released.

3 Study 1a: The cross-domain reliability landscape

3.1 Method (summary)

Study 1a systematically reviewed 217 studies across 29 domains comparing LLM-as-a-judge to human annotation. Identification triangulated OpenAlex (1500 records, 957 unique), Semantic Scholar, and a 59-query web search; screening retained on-topic empirical comparisons; 29 studies reported an extractable numeric agreement value (Figure 2; the full coded corpus is Table 5).

3.2 Findings: a green/amber/red landscape

The modal verdict is *mixed/task-dependent*; only 31% of studies sit in the reliable band and 19% in the problematic band (mean reliability 3.15/5). Reliability is governed by three properties: **objectivity**, **tractable verification**, and a **low human-agreement floor**. **Green (≥ 3.5 ; 8 domains)**: machine translation (system level), text-to-image alignment, verifiable instruction-following, QA/factuality, information extraction, general-NLG pairwise preference, dialogue, and legal scoring: here LLM judges frequently match or exceed the human ceiling (e.g., MT-Bench 85% vs. 81% human-human [48]; GEMBA-MQM 96.5% [27]; SAFE wins 76% of disagreements [45]). **Amber (19 domains)**: summarization, RAG, education, social-science annotation, medicine (structure-dependent: $\kappa \approx 0.82$ -0.93 structured vs. 0.47-0.71 complex reasoning), code, and IR/relevance (strong but contested [11]). **Red (2 domains)**: low-resource multilingual evaluation and creative writing (near-zero expert correlation [8]). Figures 3a-3b give the domain ordering and the domain×task landscape; the decision map (Figure 4b) positions domains by reliability against stakes.

Figure 1. PRISMA 2020 flow — two-arm identification (OpenAlex database + web/citation)

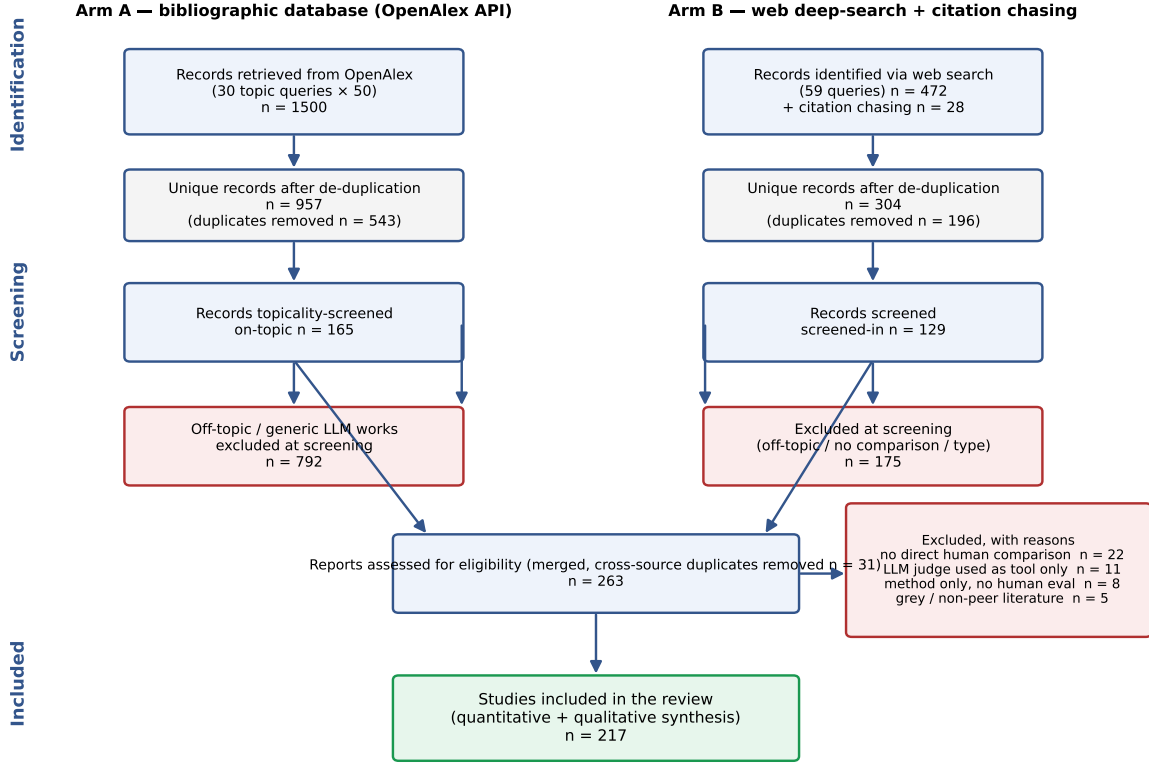


Figure 2: Study 1a PRISMA 2020 flow (two-arm identification: OpenAlex database + web/citation).

3.3 Failure modes and the decision map

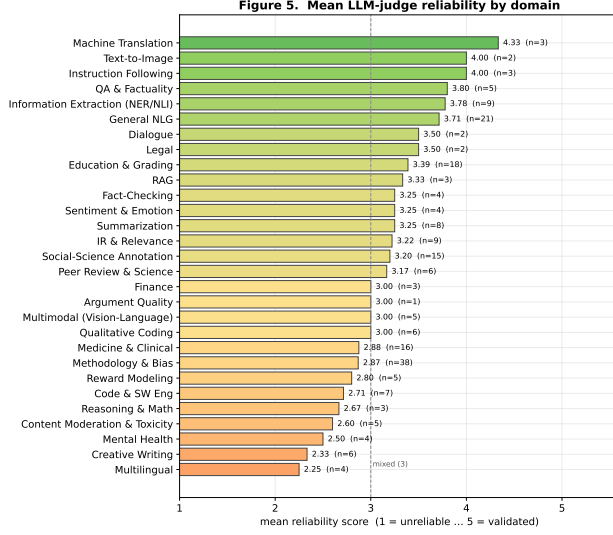
Recurring biases such as verbosity/length, position, self-preference, style-over-substance, preference leakage, non-determinism, and the *expertise gap* (judges converge to crowd, not expert, consensus [3]) all modulate reliability (Figure 4a). Agreement should be benchmarked against the human ceiling, which on subjective tasks is itself low (Krippendorff $\alpha < 0.35$ [25]). The Study-1a decision map (Figure 4b) and Table 1 translate this into deployment guidance and motivate both the moving-target test (Study 1b) and the HCC-specific analysis that follows.

4 Study 1b: Is the reliability landscape a moving target?

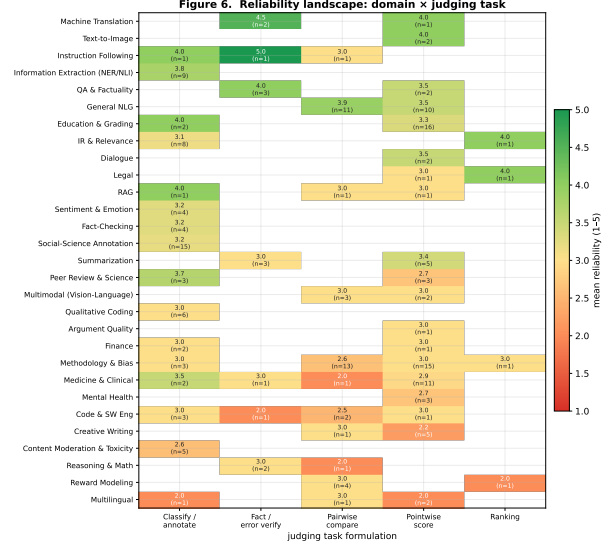
A systematic review necessarily captures results *as reported*, and most “promising”-band evidence in Study 1a was produced with proprietary judges that are now one or two model generations old (e.g. the influential G-Eval used GPT-4). This raises a question that the review alone cannot answer: when an LLM judge falls short of the human ceiling in a promising domain, is that a *temporary* limitation that more capable models will erase (“matter of time”), or a *structural* property of LLM-as-a-judge that better models do not overcome? Study 1b answers this directly for a flagship case by re-running the evaluation with the most powerful open models available to us today and comparing against the canonical reported numbers.

Table 1: Domain-level decision table. Mean reliability score (1 = unreliable ... 5 = validated), modal verdict, and traffic-light signal (GREEN ≥ 3.5 ; AMBER 2.5-3.5; RED < 2.5). Sorted by mean reliability. Small- n domains should be read together with the family-level aggregates.

Domain / task	n	Mean rel.	Modal verdict	Signal
Machine Translation	3	4.33	Promising	GREEN
Text-to-Image	2	4.00	Promising	GREEN
Instruction Following	3	4.00	Validated	GREEN
QA & Factuality	5	3.80	Promising	GREEN
Information Extraction (NER/NLI)	9	3.78	Promising	GREEN
General NLG	21	3.71	Promising	GREEN
Dialogue	2	3.50	Promising	GREEN
Legal	2	3.50	Promising	GREEN
Education & Grading	18	3.39	Mixed	AMBER
RAG	3	3.33	Mixed	AMBER
Summarization	8	3.25	Promising	AMBER
Fact-Checking	4	3.25	Promising	AMBER
Sentiment & Emotion	4	3.25	Mixed	AMBER
IR & Relevance	9	3.22	Promising	AMBER
Social-Science Annotation	15	3.20	Mixed	AMBER
Peer Review & Science	6	3.17	Mixed	AMBER
Multimodal (Vision-Language)	5	3.00	Mixed	AMBER
Qualitative Coding	6	3.00	Mixed	AMBER
Argument Quality	1	3.00	Mixed	AMBER
Finance	3	3.00	Mixed	AMBER
Medicine & Clinical	16	2.88	Mixed	AMBER
Methodology & Bias	38	2.87	Mixed	AMBER
Reward Modeling	5	2.80	Mixed	AMBER
Code & SW Eng	7	2.71	Mixed	AMBER
Reasoning & Math	3	2.67	Mixed	AMBER
Content Moderation & Toxicity	5	2.60	Mixed	AMBER
Mental Health	4	2.50	Mixed	AMBER
Creative Writing	6	2.33	Mixed	RED
Multilingual	4	2.25	Caution	RED



(a)



(b)

Figure 3: Study 1a. (a) Mean reliability by domain (green/amber/red). (b) Reliability landscape: domain×judging task; verification/classification columns are greener than open-ended scoring.

Table 2: Study 1b. Summary-level Spearman ρ of the most powerful open judges on SummEval vs. the reported G-Eval-4/GPT-4 correlations (25,600 judgments). Green cells *surpass* the reported per-aspect number.

Judge	Coher.	Consist.	Fluency	Relev.	Avg ρ
<i>Reported (G-Eval-4)</i>	0.582	0.507	0.455	0.547	0.523
deepseek-v4-pro	0.548	0.688	0.392	0.540	0.542
qwen3.6-35b-a3b	0.606	0.702	0.298	0.540	0.536
glm-5.1	0.650	0.693	0.217	0.543	0.526
gptoss-120b	0.586	0.685	0.260	0.482	0.503

4.1 Method

We select **SummEval** [16] (summarization quality assessment, a central amber-band task in Study 1a) because it pairs (i) a widely cited LLM-judge result to beat (G-Eval/GPT-4; 31) with (ii) public expert human ratings on four aspects (coherence, consistency, fluency, relevance) for $100 \times 16 = 1600$ machine summaries. We replicate the G-Eval summary-level protocol *verbatim* (same aspect definitions, 1-5 integer ratings, temperature 0) but swap the judge for the 4 most powerful open models in the Clemson catalogue (deepseek-v4-pro, glm-5.1, gptoss-120b, qwen3.6-35b-a3b). The comparison metric is the canonical *summary-level Spearman*: for each source document, correlate the judge’s 16 scores with the human means, averaged across documents. The bar to beat is G-Eval/GPT-4’s reported per-aspect correlations (coherence .582, consistency .507, fluency .455, relevance .547; average $\approx .52$). In total Study 1b contributes 25,600 fresh judgments.

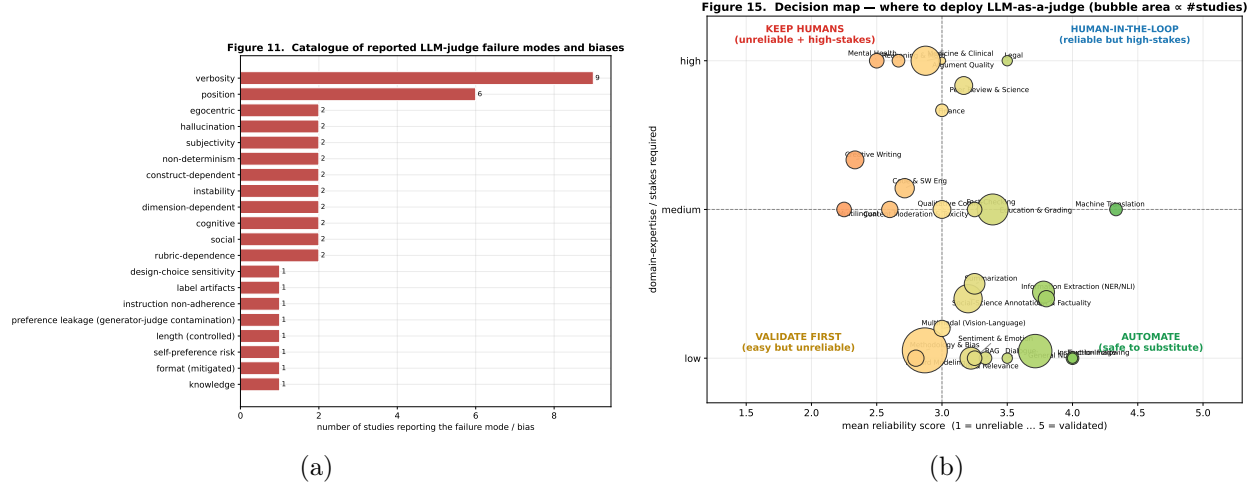


Figure 4: Study 1a. (a) Catalogue of reported failure modes/biases. (b) Deployment decision map (reliability $\times$ stakes; bubble area $\propto$ studies).

4.2 Results

Table 2 reports each open model’s summary-level Spearman against the reported G-Eval bar, and Figure 5 plots them per aspect. The most powerful open models now *match or exceed* the reported proprietary-judge correlations on SummEval. 3 of the 4 open judges beat G-Eval’s reported *average* correlation, led by **deepseek-v4-pro** (avg summary-level $\rho = 0.542$ vs. the reported ≈ 0.52). The aspect pattern is itself the finding. On **consistency** (the most *verifiable* judgment, groundable in the source document) *all four* models leap past the reported value (from .507 to .685-.702), and on **coherence** three of four do (up to .650 vs. .582); **relevance** is on par (.48-.54 vs. .55). But on **fluency** every open model falls well *short* (.22-.39 vs. a reported .455): human fluency ratings of competent machine summaries are compressed near the top of the scale (low variance), so the summary-level correlation is structurally low *regardless of how capable the judge is*, a limit that lives in the human reference, not the model, and that scale does not move.

Interpretation: time vs. structure. Study 1b shows the reliability landscape is *partly* a moving target. Where Study 1a’s amber rating reflected a capability gap on a verifiable sub-judgment, today’s open models close or exceed it; reliability there is largely a matter of time and access. But where the shortfall reflects a *compressed or contested human ceiling* (e.g. fluency, and, as Study 3 will show, subjective social annotation), even the most powerful models do not break through, because the limit lives in the human reference, not the model. This distinction, temporal capability gap vs. structural ceiling, is the lens we carry into HCC.

Takeaway for HCC. Study 1a says LLM judges are safe where the task is objective and verifiable, and Study 1b shows that frontier open models are now strong enough to match proprietary judges on exactly those verifiable tasks. But HCC’s signature methods (qualitative interpretation, participant study, design judgment) are precisely the *subjective, expert, interpretive* tasks Study 1a flags as risky and Study 1b predicts scale will *not* fix. Study 2 tests whether that prediction holds in the HCC literature itself.

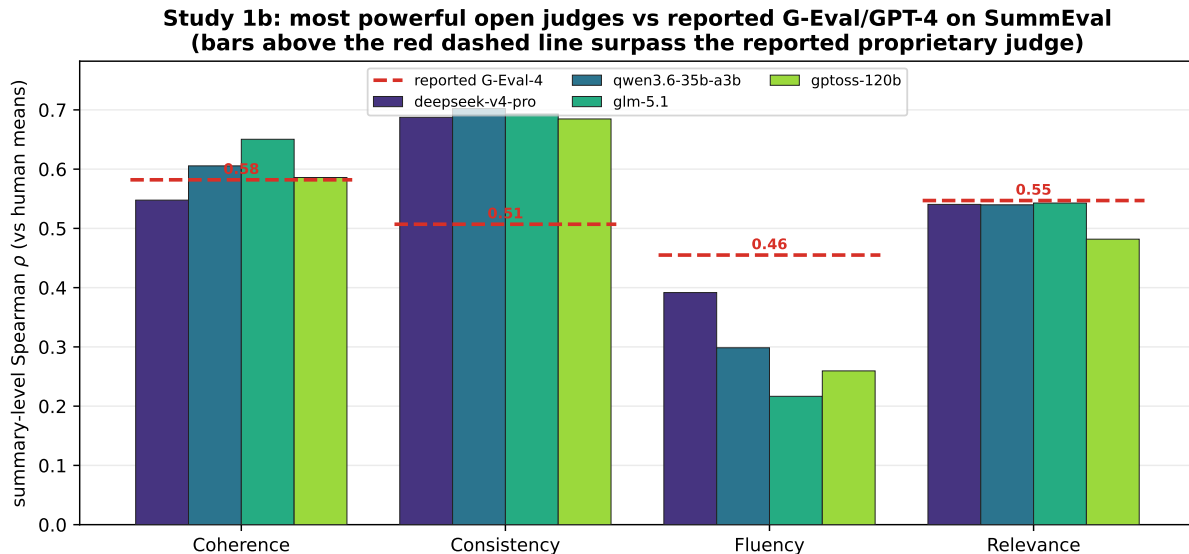


Figure 5: Study 1b. Summary-level Spearman of the most powerful open judges on SummEval, per aspect, against the reported G-Eval/GPT-4 bar (dashed). Bars above the dashed line indicate an open model that *surpasses* the reported proprietary-judge correlation.

5 Study 2: LLM substitution in HCC, HCI, and qualitative methods

5.1 Motivation and research questions

Study 1’s map is domain-general; HCC needs a method-specific account. Human-centered research is built on methods that historically *require* human input: qualitative coding, thematic and content analysis, interviews and open-ended survey analysis, crowdsourced annotation, participant studies, persona development, usability/heuristic evaluation, and design critique. Study 2 asks: **(RQ2.1)** which of these methods have been attempted with LLMs, and how reliably do LLM outputs agree with human input? **(RQ2.2)** How does the field position the LLM, to *augment*, *replace*, or *simulate* the human? **(RQ2.3)** What epistemic stance (enabling, tension, critical) does the literature take, and what does that imply for HCC practice?

5.2 Method

Following the shared protocol, we ran a Study-2-specific identification arm: 25 HCC/HCI/qualitative concept-block queries against OpenAlex (857 unique works), triangulated with the web arm and with the venue-scoped sweep of Study 1 (CHI, CSCW, UIST, IUI, TOCHI). After topicality screening and eligibility assessment we included 55 studies (Figure 6). We extracted, per study: the HCC **method**; the LLM **role** (replace / augment / simulate); reported agreement and the human baseline; the reliability **verdict**; the **epistemic stance** (enabling / tension / critical); and the **community** (HCI, CSCW, qualitative-methods, computational social science, NLP, health). The full corpus is Table 6.

5.3 RQ2.1: Which methods, and how reliable?

LLMs have been compared against human input across 15 distinct HCC methods (Figure 7a), most heavily in *thematic analysis*, *synthetic participants*, and *crowdsourced annotation*. Reliability is strongly method-dependent (Figure 7b; Table 3). The *same objectivity/interpretation axis* from

Figure S2-1. Study 2 identification (HCC/HCI/qualitative methods)

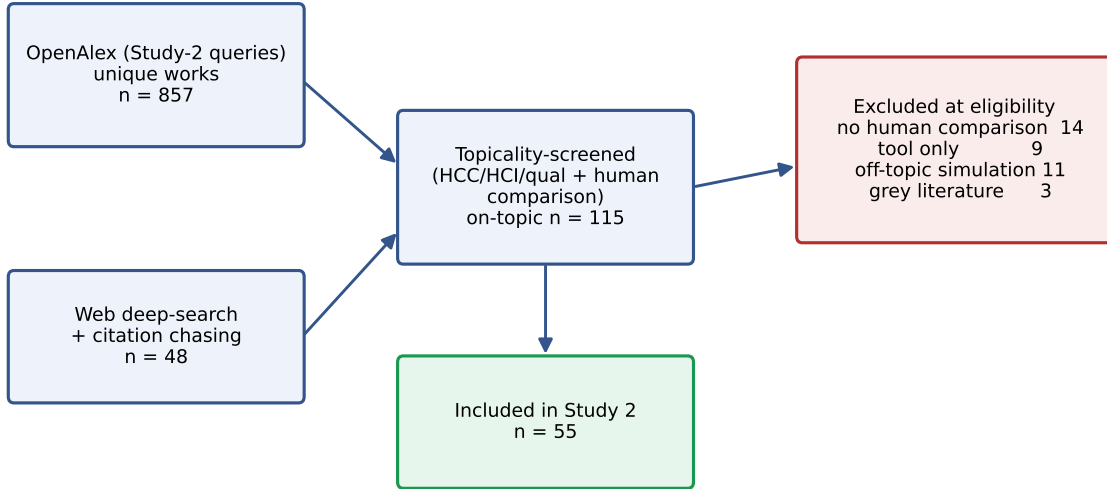


Figure 6: Study 2 identification flow (HCC/HCI/qualitative methods).

Study 1 governs HCC: **objective crowd-annotation** is the most reliable method to substitute: ChatGPT matches or beats crowd workers and even trained coders on stance, topic, and frame labeling [18, 41]; **deductive/inductive coding and content/thematic analysis** are reliable *as augmentation*, with human-equivalent agreement on structured codebooks (GPT-4 exceeding substantial κ on 8/9 hermeneutic tasks [15]; Gemini $\kappa = 0.907$ in multi-LLM thematic analysis [24]; 79.7% agreement in health thematic summarization [7]) but only moderate agreement in open interpretive settings ($\kappa \approx 0.40$ - 0.46 [30, 47]); and **participant/persona simulation and usability evaluation** are the least reliable.

5.4 RQ2.2: Augment, replace, or simulate?

HCC overwhelmingly positions LLMs to *augment* humans, not replace them: 36 of 55 studies frame the LLM as a collaborator/assistant, versus 14 as a replacement and 5 as a human simulator (Figure 8a). This is not merely rhetorical caution; it tracks reliability (Figure 9): augmentation studies skew reliable, whereas *replacement* is validated only for objective annotation and *simulation* of participants is the least reliable role (mean reliability augment 3.17 vs. replace 3.07 vs. simulate 2.8). The replacement mean is buoyed by objective crowd-annotation; remove it and replacement of *interpretive* human input is firmly amber/red. The canonical HCC workflow that emerges is human-in-the-loop: LLM proposes, human vets and retains interpretive authority [12, 17, 32].

5.5 RQ2.3: Epistemic stance and the limits of substitution

The dominant stance is *tension* (38/55), with 8 explicitly *critical* and only 9 unreservedly *enabling* (Figure 8b). Three threads recur. (i) **Interpretation and reflexivity**. Qualitative scholars warn that LLMs perform surface-level extraction well but cannot supply the situated, reflexive interpretation that gives qualitative findings their validity, and that “conditional trust” in tool output risks hollowing out analytic rigor [13, 33, 34, 40]. (ii) **Participants and identity**. Replacing human participants with LLMs *flattens and misportrays* demographic groups, positionality cannot be simulated [44], and LLM “subgroups” are often caricatures [10]; synthetic survey data distort

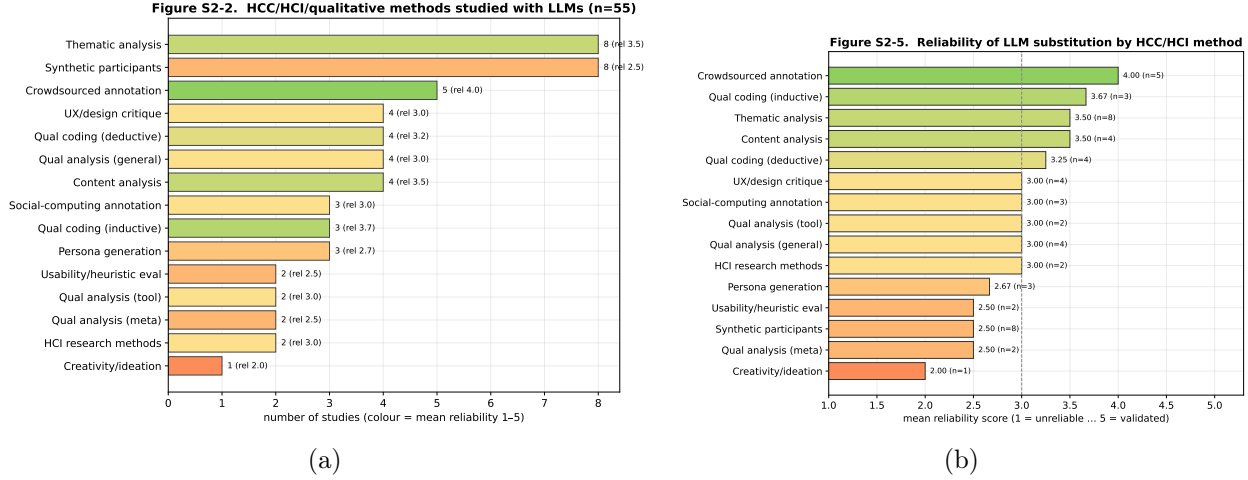


Figure 7: Study 2. (a) HCC/HCI/qualitative methods studied with LLMs (colour = mean reliability). (b) Mean reliability of LLM substitution by method.

variances and correlations [4], and an LLM is worth only a bounded number of real respondents [23]. HCI critiques accordingly recast synthetic users as “objects of inquiry,” not stand-ins [22, 39]. (iii) **Expert design judgment.** GPT-4o overlaps only 21% with expert-found usability issues and hallucinates false positives [20], and LLM UI critique trails expert designers [14], though synthetic heuristic evaluation can exceed *novice* evaluators on issue recall [49]. Figure 10 shows coverage skews to HCI and computational social science, with the augment-replace reliability gap visible across communities.

5.6 Study 2 synthesis

For HCC research, the evidence supports a clear, conditional posture. **Validated to substitute:** objective crowd/data annotation (stance, topic, relevance, frames) where a human ceiling is modest and verifiable. **Reliable only as augmentation (human-in-the-loop):** deductive/inductive qualitative coding and thematic/ content analysis: LLMs accelerate breadth and first-pass coding, but humans must own interpretation, theme development, and reflexive accounting. **Unsafe to substitute:** simulating human participants, generating personas as stand-ins for real users, and replacing expert usability/design judgment: these flatten identity, fabricate issues, and forfeit the situated knowledge that defines human-centered inquiry. This posture both confirms Study 1’s objective/subjective prediction *within* HCC and exposes concrete gaps (§6) where HCC datasets with human labels have never been formally tested with an LLM judge.

6 Study 3: An empirical LLM-as-a-judge evaluation on HCC datasets with human ground truth

Studies 1-2 expose a recurring gap: many human-centered datasets ship with human labels yet have never had a *formal* LLM-as-a-judge evaluation, meaning a panel of models, bias controls, agreement benchmarked against the human ceiling, and a contamination check. Study 3 closes this gap empirically with a released, reproducible harness.

Table 3: Study 2 method-level decision table: study count, modal LLM role, mean reliability, and traffic-light signal (GREEN ≥ 3.5 ; AMBER 2.5-3.5; RED < 2.5).

Method	n	Modal role	Mean rel.	Signal
Crowdsourced annotation	5	replace	4.00	GREEN
Qual coding (inductive)	3	augment	3.67	GREEN
Content analysis	4	augment	3.50	GREEN
Thematic analysis	8	augment	3.50	GREEN
Qual coding (deductive)	4	augment	3.25	AMBER
HCI research methods	2	augment	3.00	AMBER
Qual analysis (general)	4	augment	3.00	AMBER
Qual analysis (tool)	2	augment	3.00	AMBER
Social-computing annotation	3	replace	3.00	AMBER
UX/design critique	4	augment	3.00	AMBER
Persona generation	3	augment	2.67	AMBER
Qual analysis (meta)	2	augment	2.50	AMBER
Synthetic participants	8	simulate	2.50	AMBER
Usability/heuristic eval	2	replace	2.50	AMBER
Creativity/ideation	1	augment	2.00	RED

6.1 Method

We evaluate a panel of 6 open models served via **Clemson RCD’s vLLM API** (OpenAI-compatible), spanning five families and 9B-754B parameters (g1m-5.1 754B, gptoss-120b, deepseek-v4, gemma-4-31b, qwen3.6-27b, qwen3.5-9b), run in parallel at each model’s documented concurrency. Each model judges every item as a *virtual annotator* under the same instructions the humans received. We benchmark against the *human-human ceiling* (random-annotator-vs-majority accuracy and pairwise agreement, computed from the per-item annotator counts), report accuracy, Cohen’s κ , and macro-F1 against the majority gold, compute a count-based **alt-test** winning rate [5], aggregate the panel into a majority **jury** [42], and run a verbatim quote-completion **contamination probe**. Sample size is power-justified at $n=500$ items/dataset (95% CI half-width $\leq \pm 4.4\%$; $>80\%$ power for a 0.10 gap vs the ceiling). The run comprised **18000 judgments** across 6 datasets with *zero* API errors and a 100% answer-parse rate; only item-level labels and aggregates are released (no raw text), and the two sensitive sets are used under their public licenses.

6.2 Results

Figures 11-12 and Table 4 yield one organizing finding: **the human-agreement ceiling governs LLM reliability**. Where humans agree little (NLI; ceiling 0.64-0.75) the panel *matches or exceeds* them: on ChaosNLI-SNLI 5/6 models are **Validated** (jury 0.858 $>$ ceiling 0.752), and ChaosNLI-MNLI is **Promising**. Where humans agree strongly on a subjective social construct, the panel falls short: GoEmotions 4-way sentiment (ceiling 0.78) is uniformly **Mixed** (best 0.61); HateXplain 3-way hate (ceiling 0.835) is **Mixed** at best and **Unreliable** for the small model; binary offensiveness (SBIC, ceiling 0.905) is the strongest social-computing case (**Promising**; g1m-5.1 0.856) yet still below the ceiling. The sixth dataset is a deliberate *clean condition*: MultiPICO irony detection [6], a 2024 perspectivist corpus recent enough to be unlikely in pretraining. Its human ceiling is itself low (0.82, just above the 76% majority baseline); the panel *matches* it (**Promising**; best 0.82, jury 0.83) but largely by defaulting to the majority class (macro-F1 ≈ 0.45), and qwen3.5-9b collapses (0.40, **Unreliable**). Four further effects: (i) **granularity**: binary offensiveness (Promising) clearly

Figure S2-3. How HCC research positions LLMs: augment vs replace, and epistemic stance
(a) Intended role of the LLM **(b) Epistemic stance toward substitution**

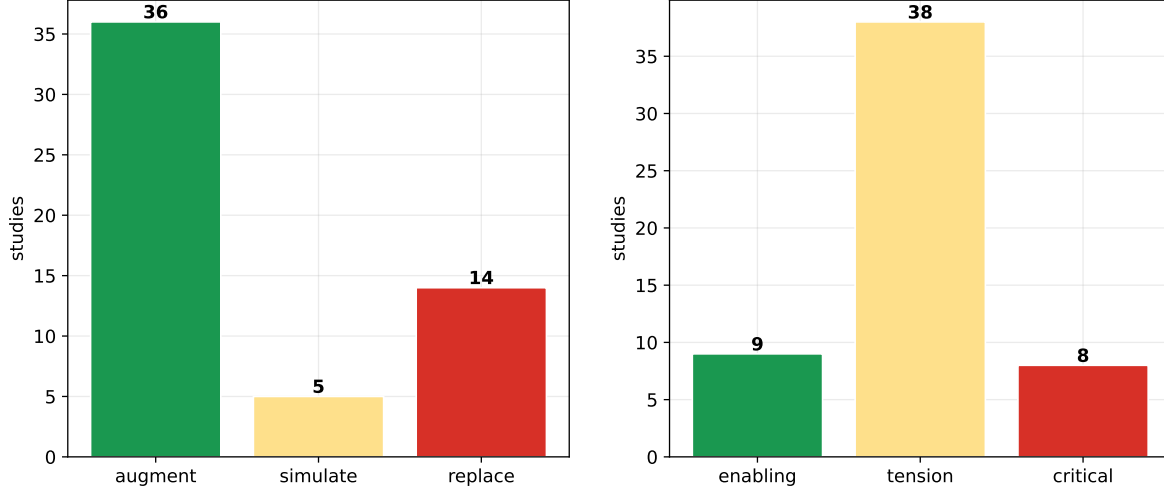


Figure 8: Study 2. (a) Intended LLM role across the HCC corpus; augmentation dominates. (b) Epistemic stance toward substitution; *tension* dominates.

Table 4: Study 3 results across 6 datasets $\times$ 6 models (18000 judgments). Human ceiling = random-annotator-vs-majority accuracy; alt-test pass = winning rate ≥ 0.5 ; contamination tier from the verbatim-recall probe.

Dataset	Construct	Human ceiling	Best model (acc)	Jury (acc)	Modal verdict	alt-test pass	Contam. tier
ChaosNLI-SNLI	NLI	0.75	0.84 (gemma-4-31b)	0.86	Validated	5/6	MEDIUM
ChaosNLI-MNLI	NLI	0.64	0.70 (gemma-4-31b)	0.65	Promising	1/6	MEDIUM
GoEmotions	emotion (4-way)	0.78	0.61 (qwen3.6-27b)	0.61	Mixed	0/6	MEDIUM
SBIC	offensive (binary)	0.91	0.86 (glm-5.1)	0.84	Promising	0/6	MEDIUM
HateXplain	hate (3-way)	0.83	0.66 (gemma-4-31b)	0.62	Mixed	0/6	MEDIUM
MultiPICO (clean)	irony (binary)	0.82	0.82 (gemma-4-31b)	0.83	Promising	5/6	MEDIUM

beats 3-way hate (Mixed); (ii) **scale**: **qwen3.5-9b** is weakest on *every* dataset (Caution/Unreliable), while 27B-754B cluster together (bigger is not always better: 31B **gemma** ties the 754B **glm**); (iii) the **alt-test** (winning rate ≥ 0.5) is passed only on the *low-ceiling* tasks, NLI and the binary irony set (5/6 models on both ChaosNLI-SNLI and MultiPICO); on every *high-consensus* task, including the Promising binary offensiveness (SBIC, 0/6), no model earns the statistical right to replace a human annotator; (iv) **juries** raise accuracy but do not lift any subjective task above its human ceiling. The **contamination probe** (Table 4, last column) shows the strongest judges have non-trivial verbatim recall across sets (gemma MEDIUM throughout), caveating the NLI “Validated” result; crucially the recent low-contamination MultiPICO (LOW for the small-recall **qwen3.5-9b**) is still matched only at its low human ceiling, so the social-computing shortfall tracks the human ceiling, not memorization.

6.3 What governs reliability? A mixed-effects analysis

To test the “ceiling governs” claim inferentially rather than descriptively, we model every individual judgment ($N=18000$; outcome = $\mathbf{1}[\text{judge} = \text{majority gold}]$) with a logistic specification whose fixed

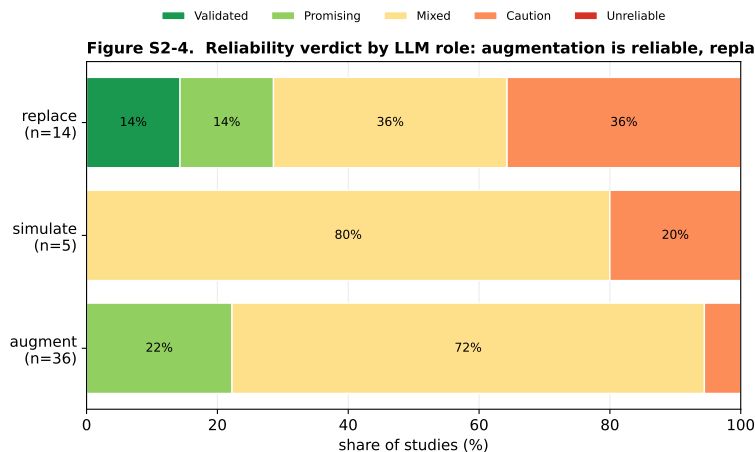


Figure 9: Study 2. Reliability verdict by LLM role. Augmentation is reliable; replacement is reliable only for objective annotation; participant/usability *simulation* is the least reliable.

effects are task *subjectivity* (1–human-ceiling accuracy, at the dataset level) and model *scale* ($\log_{10}$ parameters), reported both as a Bayesian mixed-effects logistic GLMM with random intercepts for dataset and item, and, for clean inference, as a logistic GLM with item-cluster-robust standard errors (Figure 13). Three results stand out. (i) **Subjectivity dominates:** each unit of 1–ceiling multiplies the odds of agreeing with humans by $OR = 0.17$ ($p = 1e - 7$). (ii) **Scale helps, but modestly:** larger models do better ($OR = 1.30$ per $\log_{10}$ params, $z = 13.40$), an effect far smaller than subjectivity’s, consistent with Study 1b’s “partly a moving target.” (iii) **Variance lives in the items:** the GLMM random-intercept standard deviation is far larger across *items* ($SD = 2.02$) than across datasets ($SD = 0.73$), i.e. *which item* is being judged, its latent human (dis)agreement, explains judge correctness better than which model judges or even which dataset it comes from. Finally, a Benjamini-Hochberg analysis across all 36 judge×dataset cells finds 26 significant at $q < .05$: **22 sit significantly below the human ceiling** and only 4 significantly above (the objective-NLI cases). Scaling the judge shifts cells modestly rightward; it does not move a high-ceiling subjective task across the line.

6.4 Integration with the decision framework

Mapped onto the five-level rubric, Study 3 drops fresh, dataset-level points into the Study-1/2 landscape and *quantitatively confirms* the augment-not-replace conclusion for human-centered annotation: LLM judges are a validated substitute for objective/low-ceiling labeling and a strong assistant for coarse (binary) moderation, but fall short of, and fail the replacement test for, subjective, fine-grained, or high-consensus human-centered constructs, a pattern the mixed-effects model shows is driven by item-level human (dis)agreement rather than by model scale. The recent MultiPICo clean condition rules out contamination as the explanation: a 2024 set the low-recall judge has not memorized is still matched only at its (low) human ceiling. *Limitation:* the panel is limited to open models from a single provider, and qualitative-coding corpora with public multi-coder labels (still scarce) are the immediate next addition.

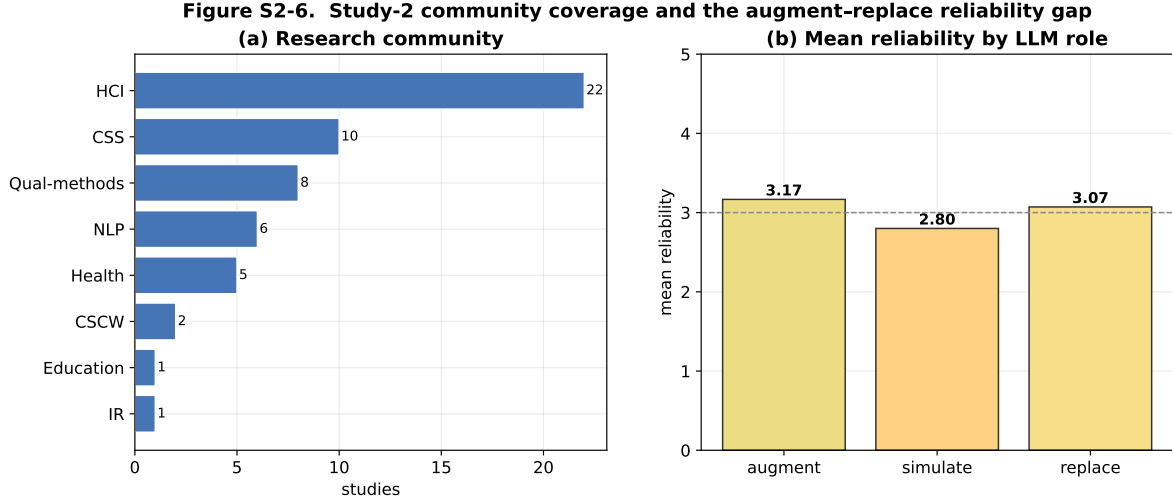


Figure 10: Study 2. (a) Coverage by research community. (b) Mean reliability by LLM role, augmentation > replacement > simulation.

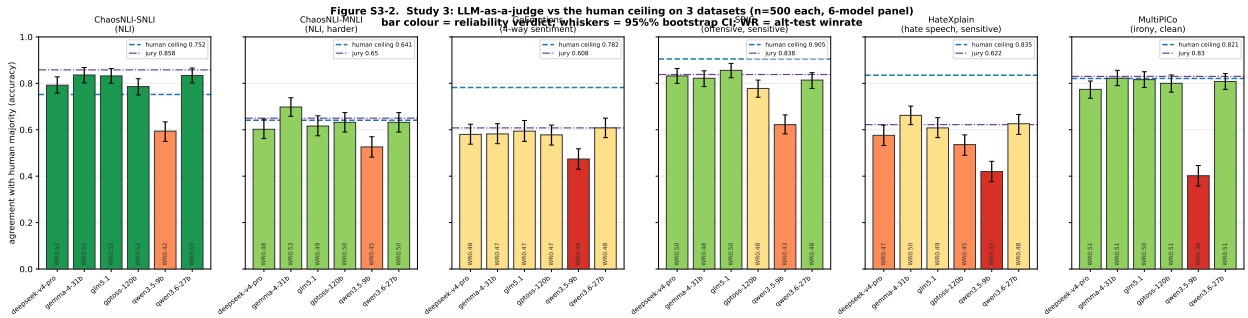


Figure 11: Study 3. Per-dataset accuracy vs the human ceiling (blue dashed) for the 6-model panel (n=500 each; whiskers = 95% bootstrap CI; bar colour = verdict; WR = alt-test winning rate). Datasets ordered by construct subjectivity. LLMs clear LOW ceilings (NLI) but fall short of HIGH ceilings on subjective social-computing tasks.

7 General discussion: a decision framework for HCC

Across both reviews, one principle organizes the evidence: **LLM-as-a-judge substitution is reliable to the degree that the human task is objective, verifiable, low in required expertise, and benchmarked against a modest human ceiling**, and it degrades as tasks become interpretive, expert, identity-laden, or open-ended. For Human-Centered Computing specifically:

- **Automate (with light audit):** objective annotation/labeling of user-generated content such as stance, topic, relevance, frames, and simple sentiment [18, 41].
- **Augment, human-in-the-loop:** qualitative coding and thematic/content analysis: use the LLM for first-pass coding, breadth, and disagreement surfacing, but keep human interpretive authority, calibrate on a labeled subset, and report reliability against human coders [17, 32, 36].
- **Keep humans (do not substitute):** simulating participants or replacing them with personas,

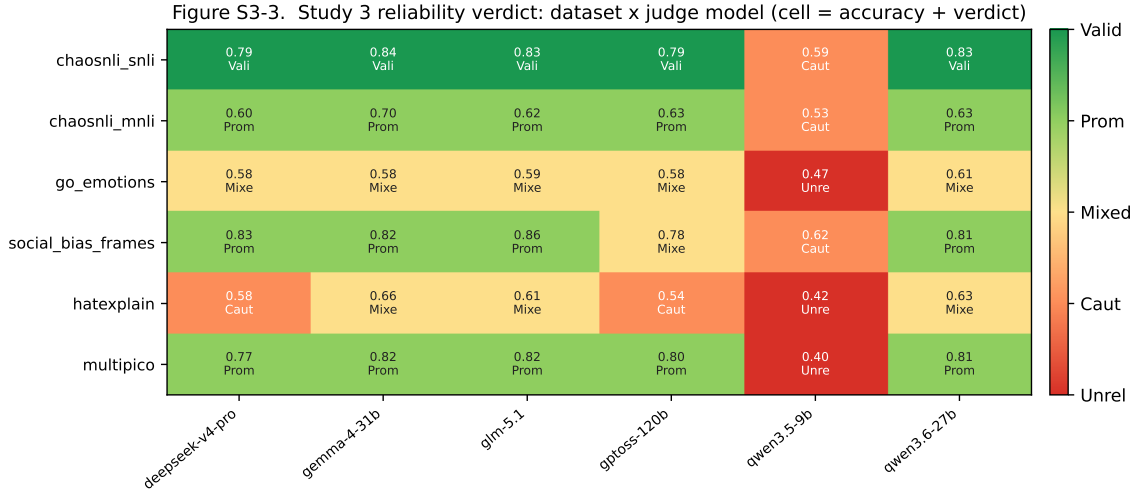


Figure 12: Study 3. Dataset×judge-model reliability verdict (cell = accuracy + verdict). The scale effect (**qwen3.5-9b** column) and the construct-subjectivity gradient (rows) are both visible.

and replacing expert usability/design judgment: these flatten identity, fabricate evidence, and abandon situated knowledge [20, 39, 44].

Practically, HCC researchers reporting LLM-assisted analysis should: state the LLM’s *role* (augment/replace/simulate); report agreement against a *human-human* baseline at the right granularity; control position/length bias and pin model/version; prefer comparative scoring and juries to single absolute judges; and treat the alt-test [5] as the bar for any replacement claim.

8 Limitations

Both reviews used a single-reviewer, LLM-assisted pipeline (logs released) rather than dual human reviewers; metrics are heterogeneous and synthesized within families rather than pooled; several methods/domains rest on small n (flagged via family/role aggregates); database recall depends on indexing and query phrasing (mitigated, not removed, by three-source triangulation); and findings reflect models available through mid-2026. The empirical studies are bounded too: Study 1b uses one flagship promising-domain benchmark (SummEval), and Study 3’s panel is limited to open models from a single provider and to 6 datasets (one of which, MultiPICO, is included as a recent low-contamination clean condition), with qualitative-coding corpora still in progress. The verdict rubric, though explicit and released, embeds judgement; we therefore release both corpora for re-coding.

9 Conclusion

LLMs are already remaking how Human-Centered Computing research is done. Our four studies show that this is neither to be embraced nor refused wholesale. Across 217 cross-domain studies and 55 HCC-specific studies, the evidence converges on a conditional verdict: an LLM-as-a-judge is a *validated substitute* for objective annotation, a *valuable human-in-the-loop assistant* for qualitative coding and analysis, and an *unsafe replacement* for human participants, personas, and expert design judgment. Study 1b shows that the most powerful open models now match or exceed proprietary judges on the verifiable parts of “promising” tasks, and Study 3 turns the remaining “mixed” verdicts

Figure S3-4. Study 3 inferential layer: what governs LLM-judge agreement

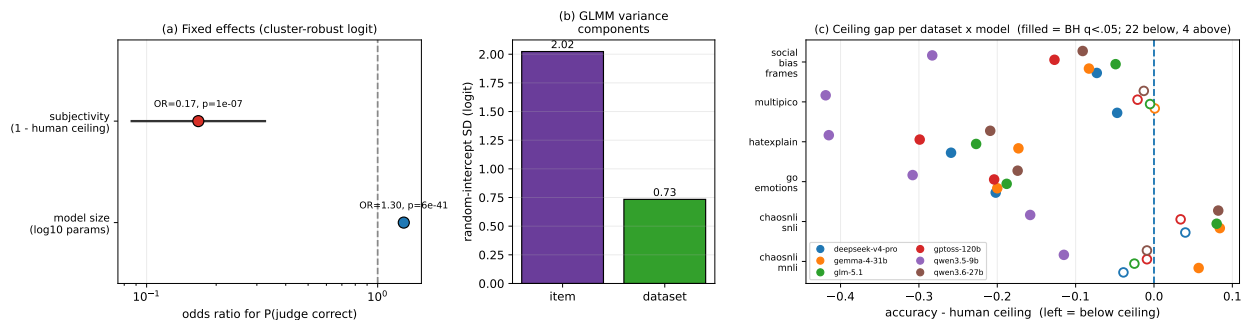


Figure 13: Study 3 inferential layer. (a) Fixed-effect odds ratios (item-cluster-robust logit): subjectivity is a strong negative predictor of judge correctness, scale a weak positive one. (b) GLMM variance components: item-level variance dominates. (c) Ceiling gap per judge x dataset (filled = BH $q < .05$); most cells sit below the human ceiling.

into dataset-level evidence using open, locally-servable models and the field’s own reliability standards. The right question for HCC is never “can a machine judge?” but “*can this machine replace a second human, on this method, at this granularity, under these controls?*”, and this paper makes that question answerable, method by method.

Reproducibility

Two coded corpora (Study 1: 217 studies; Study 2: 55 studies), three search logs, PRISMA counts, all figure/table-generation code, the auto-generated statistics, and the Study-3 evaluation harness are released. Everything regenerates via `make all`.

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A Study 1 corpus

Table 5: Full extracted-studies corpus (217 included studies). Agreement is the headline agreement/percentage/percentage of proportion where reported.

Key	Study	Domain	Task	Metri
mirzakhmedova2024argument	Mirzakhmedova et al. (2024)	Argument Quality	pointwise	pct_
tong2024codejudge	Tong et al. (2024)	Code & SW Eng	classify	accu
watson2024aicode	Watson et al. (2024)	Code & SW Eng	classify	pct_
codespec2025	Jin et al. (2025)	Code & SW Eng	verify	accu
wang2025setesting	Wang et al. (2025)	Code & SW Eng	pairwise	pct_
zhuo2025codejudgeeff	Wang et al. (2025)	Code & SW Eng	pointwise	kapp
biasloop2026	Zhao et al. (2026)	Code & SW Eng	pairwise	pct_
overcorrect2026	Jin et al. (2026)	Code & SW Eng	classify	accu
hatespeechprobe2023	Roy et al. (2023)	Content Moderation & Toxicity	classify	f1
li2023hotchatgpt	Li et al. (2023)	Content Moderation & Toxicity	classify	pct_
nguyen2024crisismisinfo	Nguyen & Rudra (2024)	Content Moderation & Toxicity	classify	f1
hatespeechanno2025	authors (2025)	Content Moderation & Toxicity	classify	kapp
icwsm2025hatebias	Giorgi et al. (2025)	Content Moderation & Toxicity	classify	kapp
xie2023nextchapter	Xie et al. (2023)	Creative Writing	pointwise	pear
chakrabarty2024artartifice	Chakrabarty et al. (2024)	Creative Writing	pointwise	pear
ismayilzada2024creative	Ismayilzada et al. (2024)	Creative Writing	pointwise	pct_

Key	Study	Domain	Task	Metri
creativeevalmdpi2025	Kim et al. (2025)	Creative Writing	pointwise	spea
haase2025creativitypeaked	Haase et al. (2025)	Creative Writing	pointwise	pct_
litbench2025	Fein et al. (2025)	Creative Writing	pairwise	pct_
lin2023llmeval	Lin & Chen (2023)	Dialogue	pointwise	spea
mendonca2024dialogue	Mendonca et al. (2024)	Dialogue	pointwise	spea
hackl2023reliablerater	Hackl et al. (2023)	Education & Grading	pointwise	icc
khademi2023bardassessment	Khademi (2023)	Education & Grading	pointwise	pct_
kortemeyer2023physics	Kortemeyer (2023)	Education & Grading	pointwise	pct_
mizumoto2023aes	Mizumoto & Eguchi (2023)	Education & Grading	pointwise	qwk
yancey2023cefr	Yancey et al. (2023)	Education & Grading	pointwise	qwk
aescomparative2024	Lee et al. (2024)	Education & Grading	pointwise	qwk
floden2024gradingexams	Flodén (2024)	Education & Grading	pointwise	pct_
grevisse2024asagmed	Grévisse (2024)	Education & Grading	classify	pct_
hou2024classroom	Hou et al. (2024)	Education & Grading	pointwise	icc
k12shortanswer2024	Henkel et al. (2024)	Education & Grading	classify	kapp
koraishi2024langassess	Koraishi (2024)	Education & Grading	pointwise	pct_
quah2024dentalaes	Quah et al. (2024)	Education & Grading	pointwise	icc
wang2024beyondagreement	Wang et al. (2024)	Education & Grading	pointwise	qwk
writingmultidim2024	Steiss et al. (2024)	Education & Grading	pointwise	qwk
asagfair2025	Henkel et al. (2025)	Education & Grading	pointwise	kapp
sefl2025feedback	authors (2025)	Education & Grading	pointwise	fl
studentsjudge2025	Banihashem et al. (2025)	Education & Grading	pointwise	pct_
k12sci2026	He et al. (2026)	Education & Grading	pointwise	pct_
claimmatch2023	Choi et al. (2023)	Fact-Checking	classify	pct_
ni2024afacta	Ni et al. (2024)	Fact-Checking	classify	pct_
politicaltruths2024	Chatrath et al. (2024)	Fact-Checking	classify	pct_
factsopinions2025	Saju et al. (2025)	Fact-Checking	classify	accu
kaikaus2023financecorpora	Kaikaus et al. (2023)	Finance	classify	accu
bizfinbench2025	Lu et al. (2025)	Finance	pointwise	pct_
trident2025	Hui et al. (2025)	Finance	classify	pct_
chan2023chateval	Chan et al. (2023)	General NLG	pairwise	pct_
chiang2023alt	Chiang & Lee (2023)	General NLG	pointwise	pct_
dubois2023alpacafarm	Dubois et al. (2023)	General NLG	pairwise	winn
hsu2023figurecaption	Hsu et al. (2023)	General NLG	pointwise	spea
kim2023prometheus	Kim et al. (2023)	General NLG	pointwise	pear
lai2023styletransfer	Lai et al. (2023)	General NLG	pointwise	spea
li2023autoj	Li et al. (2023)	General NLG	pairwise	pct_
wang2023chatgptnlg	Wang et al. (2023)	General NLG	pointwise	spea
wang2023pandalm	Wang et al. (2023)	General NLG	pairwise	accu
xu2023instructscore	Xu et al. (2023)	General NLG	pointwise	pear
zhang2023widerdeeper	Zhang et al. (2023)	General NLG	pairwise	pct_
zheng2023mtbench	Zheng et al. (2023)	General NLG	pairwise	pct_
chiang2024arena	Chiang et al. (2024)	General NLG	pairwise	pct_
doostmohammadi2024autoeval	Doostmohammadi et al. (2024)	General NLG	pointwise	pct_
dubois2024lc	Dubois et al. (2024)	General NLG	pairwise	winn
hashemi2024llmrubric	Hashemi et al. (2024)	General NLG	pointwise	pear
huang2024nlexpl	Huang et al. (2024)	General NLG	pointwise	pct_
kim2024prometheus2	Kim et al. (2024)	General NLG	pointwise	pear
li2024arenahard	Li et al. (2024)	General NLG	pairwise	pct_
lin2024wildbench	Lin et al. (2024)	General NLG	pairwise	winn
verga2024poll	Verga et al. (2024)	General NLG	pairwise	pct_
faggioli2023perspectives	Faggioli et al. (2023)	IR & Relevance	classify	kapp
alaofi2024fooledrelevant	Alaofi et al. (2024)	IR & Relevance	classify	pct_
clarke2024cantreplace	Clarke & Dietz (2024)	IR & Relevance	classify	kapp
thomas2024relevance	Thomas et al. (2024)	IR & Relevance	classify	kapp
upadhyay2024largescale	Upadhyay et al. (2024)	IR & Relevance	classify	kapp
zhuang2024beyondyesno	Zhuang et al. (2024)	IR & Relevance	ranking	kend

Key	Study	Domain	Task	Metri
arabzadeh2025benchrelevance	Arabzadeh & Clarke (2025)	IR & Relevance	classify	kapp
arabzadeh2025promptsens	Arabzadeh & Clarke (2025)	IR & Relevance	classify	kapp
frobe2025assessorsagree	Frobe et al. (2025)	IR & Relevance	classify	kapp
li2023coannotating	Li et al. (2023)	Information Extraction (NER/NLI)	classify	pct_
humanllmcollab2024	Wang et al. (2024)	Information Extraction (NER/NLI)	classify	pct_
kholodna2024activelearning	Kholodna et al. (2024)	Information Extraction (NER/NLI)	classify	fl_
kim2024meganno	Kim et al. (2024)	Information Extraction (NER/NLI)	classify	fl_
nasution2024chatgptlabel	Nasution & Onan (2024)	Information Extraction (NER/NLI)	classify	accu
ner2024augment	Naraki et al. (2024)	Information Extraction (NER/NLI)	classify	fl_
nli2024labeldist	Chen et al. (2024)	Information Extraction (NER/NLI)	classify	pear
ronningstad2024gptannotator	Ronningstad et al. (2024)	Information Extraction (NER/NLI)	classify	fl_
qiu2025labelensemble	Qiu et al. (2025)	Information Extraction (NER/NLI)	classify	accu
zhou2023ifeval	Zhou et al. (2023)	Instruction Following	verify	accu
zeng2024llmbar	Zeng et al. (2024)	Instruction Following	pairwise	accu
ifcritic2025	Wen et al. (2025)	Instruction Following	classify	pct_
legaldocrec2025	Pradhan et al. (2025)	Legal	ranking	spea
lemaj2025	Enguehard et al. (2025)	Legal	pointwise	pct_
kocmi2023gemba	Kocmi & Federmann (2023)	Machine Translation	pointwise	kend
kocmi2023gembamqm	Kocmi & Federmann (2023)	Machine Translation	verify	pct_
lu2023erroranalysis	Lu et al. (2023)	Machine Translation	verify	kend
sun2023radimpressions	Sun et al. (2023)	Medicine & Clinical	pointwise	pct_
tang2023medevidence	Tang et al. (2023)	Medicine & Clinical	pointwise	pct_
brain2024radgpt	Mitsuyama et al. (2024)	Medicine & Clinical	classify	accu
brake2024clinicalnote	Brake & Schaaf (2024)	Medicine & Clinical	pointwise	pct_
wang2024clinicalevent	Wang et al. (2024)	Medicine & Clinical	classify	pct_
chung2025verifact	Chung et al. (2025)	Medicine & Clinical	verify	pct_
clinstruct2025	Brake et al. (2025)	Medicine & Clinical	pointwise	icc
croxford2025autoeval	Croxford et al. (2025)	Medicine & Clinical	pointwise	pct_
croxford2025clinsummjudge	Croxford et al. (2025)	Medicine & Clinical	pointwise	pct_
healthbench2025	Arora et al. (OpenAI) (2025)	Medicine & Clinical	pointwise	pct_
kim2025emulate	Kim & Kim (2025)	Medicine & Clinical	pointwise	pct_
llmevalmed2025	Zhang et al. (2025)	Medicine & Clinical	pointwise	pct_
szymanski2025limits	Szymanski et al. (2025)	Medicine & Clinical	pairwise	pct_
clindialog2026	Sun et al. (2026)	Medicine & Clinical	pointwise	pct_
frenchmedqa2026	Belmadani et al. (2026)	Medicine & Clinical	pointwise	kapp
medjudge2026scoping	Li et al. (2026)	Medicine & Clinical	review	nan
counselbench2025	Li et al. (2025)	Mental Health	pointwise	pct_
hua2025mhscoping	Hua et al. (2025)	Mental Health	review	nan
mhtrust2025	Badawi et al. (2025)	Mental Health	pointwise	icc
mhsupport2026	Badawi et al. (2026)	Mental Health	pointwise	icc
liu2023trustworthy	Liu et al. (2023)	Methodology & Bias	survey	nan
saito2023verbositybias	Saito et al. (2023)	Methodology & Bias	pairwise	pct_
wang2023shepherd	Wang et al. (2023)	Methodology & Bias	pointwise	pct_
zhu2023judgelm	Zhu et al. (2023)	Methodology & Bias	pairwise	pct_
bavaresco2024judgebench	Bavaresco et al. (2024)	Methodology & Bias	classify	kapp
chen2024humansorllms	Chen et al. (2024)	Methodology & Bias	pairwise	pct_
desmond2024evalullm	Desmond et al. (2024)	Methodology & Bias	pointwise	pct_
fan2024sedareval	Fan et al. (2024)	Methodology & Bias	pointwise	pct_
feuer2024style	Feuer et al. (2024)	Methodology & Bias	pairwise	pct_
gu2024survey	Gu et al. (2024)	Methodology & Bias	survey	nan
jung2024trustescalate	Jung et al. (2024)	Methodology & Bias	pairwise	pct_
justice2024prejudice	Ye et al. (2024)	Methodology & Bias	pairwise	pct_
ke2024critiquellm	Ke et al. (2024)	Methodology & Bias	pointwise	spea
koo2024cobbler	Koo et al. (2024)	Methodology & Bias	pairwise	winr
li2024compsurvey	Li et al. (2024)	Methodology & Bias	survey	nan
li2024gen2judge	Li et al. (2024)	Methodology & Bias	survey	nan
li2024splitmerge	Li et al. (2024)	Methodology & Bias	pairwise	pct_

Key	Study	Domain	Task	Metri
liu2024pairwisepref	Liu et al. (2024)	Methodology & Bias	pairwise	pct_
pan2024hcdjudge	Pan et al. (2024)	Methodology & Bias	pointwise	nan
panickssery2024selfpref	Panickssery et al. (2024)	Methodology & Bias	pairwise	pct_
raina2024robust	Raina et al. (2024)	Methodology & Bias	pointwise	pct_
shankar2024validators	Shankar et al. (2024)	Methodology & Bias	pointwise	pct_
stureborg2024inconsistent	Stureborg et al. (2024)	Methodology & Bias	pointwise	pct_
thakur2024judgingjudges	Thakur et al. (2024)	Methodology & Bias	pairwise	pct_
wataoka2024selfpref	Wataoka et al. (2024)	Methodology & Bias	pairwise	pct_
zhu2024nationalitybias	Zhu et al. (2024)	Methodology & Bias	pointwise	pct_
calderon2025alttest	Calderon et al. (2025)	Methodology & Bias	classify	pct_
gao2025nlgsurvey	Gao et al. (2025)	Methodology & Bias	survey	nan
gao2025reranking	Gao et al. (2025)	Methodology & Bias	ranking	kend
hu2025trainjudge	Hu et al. (2025)	Methodology & Bias	pointwise	pct_
huang2025empiricaljudge	Huang et al. (2025)	Methodology & Bias	pointwise	pct_
judgesverdict2025	Han et al. (2025)	Methodology & Bias	pointwise	kapp
li2025prefleakage	Li et al. (2025)	Methodology & Bias	pairwise	pct_
murugadoss2025evaltheeval	Murugadoss et al. (2025)	Methodology & Bias	pointwise	pct_
voutsas2025biaseddesign	Voutsas et al. (2025)	Methodology & Bias	classify	pct_
yamauchi2025designchoices	Yamauchi et al. (2025)	Methodology & Bias	pointwise	pct_
imrit2026iaa	James et al. (2026)	Methodology & Bias	review	kripp
temperature2026	Li et al. (2026)	Methodology & Bias	pointwise	pct_
khondaker2023gptaraeval	Khondaker et al. (2023)	Multilingual	classify	fl
mmeval2024	Son et al. (2024)	Multilingual	pairwise	accu
multiljudge2025	Fu et al. (2025)	Multilingual	pointwise	pct_
reliablemulti2026	Zubiaga et al. (2026)	Multilingual	pointwise	pct_
chen2024mllmjudge	Chen et al. (2024)	Multimodal (Vision-Language)	pairwise	pct_
duan2024uimockup	Duan et al. (2024)	Multimodal (Vision-Language)	pointwise	pct_
lee2024promvision	Lee et al. (2024)	Multimodal (Vision-Language)	pointwise	pear
lu2024wildvision	Lu et al. (2024)	Multimodal (Vision-Language)	pairwise	pct_
yasunaga2025mmrewardbench	Yasunaga et al. (2025)	Multimodal (Vision-Language)	pairwise	accu
chatgptreviewers2024	Ho et al. (2024)	Peer Review & Science	pointwise	pct_
liang2024feedback	Liang et al. (2024)	Peer Review & Science	pointwise	pct_
murad2024quality	Tarakji et al. (2024)	Peer Review & Science	classify	pct_
prismascreen2024	Guo et al. (2024)	Peer Review & Science	classify	kapp
rose2025rob	Rose et al. (2025)	Peer Review & Science	classify	kapp
shopovski2025peerreview	Shopovski et al. (2025)	Peer Review & Science	pointwise	pct_
guo2023howclose	Guo et al. (2023)	QA & Factuality	pointwise	pct_
min2023factscore	Min et al. (2023)	QA & Factuality	verify	pct_
li2024pedants	Li et al. (2024)	QA & Factuality	verify	pct_
madaan2024refverdict	Reference-Guided Verdict authors (2024)	QA & Factuality	pointwise	pct_
wei2024safe	Wei et al. (2024)	QA & Factuality	verify	pct_
leas2023contentanalysis	Leas et al. (2023)	Qualitative Coding	classify	pct_
meng2024chalet	Meng et al. (2024)	Qualitative Coding	classify	pct_
qualigpt2024	Zhang et al. (2024)	Qualitative Coding	classify	pct_
mehta2025qualgenai	Mehta et al. (2025)	Qualitative Coding	classify	pct_
qualcoding2025jla	Xiao et al. (2025)	Qualitative Coding	classify	fleiss
thematic2025charity	Wen et al. (2025)	Qualitative Coding	classify	pct_
es2023ragas	Es et al. (2023)	RAG	pointwise	pct_
saadfalcon2023ares	Saad-Falcon et al. (2023)	RAG	classify	accu
jin2024ragreward	Jin et al. (2024)	RAG	pairwise	accu
raju2024calc2adj	Raju et al. (2024)	Reasoning & Math	verify	accu
tan2024judgebench	Tan et al. (2024)	Reasoning & Math	pairwise	accu
mathrobust2026	Yosef et al. (2026)	Reasoning & Math	verify	accu
lee2023rlaif	Lee et al. (2023)	Reward Modeling	pairwise	pct_
lambert2024rewardbench	Lambert et al. (2024)	Reward Modeling	pairwise	accu
wu2024metarewarding	Wu et al. (2024)	Reward Modeling	pairwise	pct_
yuan2024selfrewarding	Yuan et al. (2024)	Reward Modeling	pairwise	pct_

Key	Study	Domain	Task	Metri
ifreward2026	Wen et al. (2026)	Reward Modeling	ranking	kend
belal2023sentiment	Belal et al. (2023)	Sentiment & Emotion	classify	accu
koptyra2023clarinemo	Koptyra et al. (2023)	Sentiment & Emotion	classify	pct_
zhang2023sentiment	Zhang et al. (2023)	Sentiment & Emotion	classify	accu
niu2024text2emotion	Niu et al. (2024)	Sentiment & Emotion	classify	pct_
aldeen2023chatgptvshuman	Aldeen et al. (2023)	Social-Science Annotation	classify	accu
cegin2023paraphrases	Cegin et al. (2023)	Social-Science Annotation	classify	pct_
gilardi2023chatgpt	Gilardi et al. (2023)	Social-Science Annotation	classify	accu
ollion2023mindhype	Ollion et al. (2023)	Social-Science Annotation	classify	pct_
ostyakova2023crowdexperts	Ostyakova et al. (2023)	Social-Science Annotation	classify	pct_
pangakis2023validation	Pangakis et al. (2023)	Social-Science Annotation	classify	f1
reiss2023testing	Reiss (2023)	Social-Science Annotation	classify	pct_
wu2023llmworkers	Wu et al. (2023)	Social-Science Annotation	classify	pct_
zhu2023reproduce	Zhu et al. (2023)	Social-Science Annotation	classify	pct_
li2024stance	Li & Conrad (2024)	Social-Science Annotation	classify	pct_
pangakis2024keephumans	Pangakis & Wolken (2024)	Social-Science Annotation	classify	f1
ziems2024css	Ziems et al. (2024)	Social-Science Annotation	classify	accu
ahmad2025moralmachine	Ahmad & Takemoto (2025)	Social-Science Annotation	classify	pct_
horych2025promisespitfalls	Horych et al. (2025)	Social-Science Annotation	classify	f1
tornberg2025political	Törnberg (2025)	Social-Science Annotation	classify	accu
gao2023humanlike	Gao et al. (2023)	Summarization	pointwise	spea
liu2023geval	Liu et al. (2023)	Summarization	pointwise	spea
luo2023factinconsistency	Luo et al. (2023)	Summarization	verify	pct_
shen2023notyet	Shen et al. (2023)	Summarization	pointwise	spea
lee2024unisumeval	Lee et al. (2024)	Summarization	pointwise	pct_
song2024finesure	Song et al. (2024)	Summarization	verify	pct_
tang2024tofueval	Tang et al. (2024)	Summarization	verify	pct_
zhang2024newssumm	Zhang et al. (2024)	Summarization	pointwise	pct_
wu2023hpsv2	Wu et al. (2023)	Text-to-Image	pointwise	accu
lin2024vqascore	Lin et al. (2024)	Text-to-Image	pointwise	pear

B Study 2 corpus

Table 6: Study 2 corpus (55 studies): HCC/HCI/qualitative methods using LLMs vs human input.

Study	Method	Role	Community	Agree	Verdict
Chew et al. (2023)	Content analysis	augment	CSS	-	Promising
Leas et al. (2023)	Content analysis	augment	Health	-	Mixed
Long et al. (2024)	Content analysis	augment	Education	-	Mixed
Dunivin (2025)	Content analysis	augment	CSS	-	Promising
Anderson et al. (2024)	Creativity/ideation	augment	HCI	-	Caution
Gilardi et al. (2023)	Crowdsourced annotation	replace	CSS	-	Validated
Tornberg (2023)	Crowdsourced annotation	replace	CSS	-	Validated
Kuzman et al. (2023)	Crowdsourced annotation	replace	NLP	-	Mixed
Li et al. (2023)	Crowdsourced annotation	augment	NLP	-	Promising
Ostyakova et al. (2023)	Crowdsourced annotation	replace	NLP	-	Mixed
Tabone & de Winter (2023)	HCI research methods	augment	HCI	-	Mixed
Aubin Le Quere et al. (2024)	HCI research methods	augment	HCI	-	Mixed
Salminen et al. (2024)	Persona generation	augment	HCI	-	Mixed
Jung et al. (2025)	Persona generation	augment	HCI	-	Mixed
Haxvig et al. (2025)	Persona generation	replace	HCI	-	Caution
Zhang et al. (2023)	Qual analysis (general)	augment	Qual-methods	-	Mixed
Wachinger et al. (2024)	Qual analysis (general)	augment	Health	-	Mixed
Li et al. (2024)	Qual analysis (general)	augment	Health	0.40	Mixed

Study	Method	Role	Community	Agree	Verdict
Mehta et al. (2025)	Qual analysis (general)	augment	Qual-methods	-	Mixed
Davison et al. (2024)	Qual analysis (meta)	augment	CSS	-	Caution
Schroeder et al. (2025)	Qual analysis (meta)	augment	HCI	-	Mixed
Zhang et al. (2024)	Qual analysis (tool)	augment	HCI	-	Mixed
Meng et al. (2024)	Qual analysis (tool)	augment	HCI	-	Mixed
Xiao et al. (2023)	Qual coding (deductive)	augment	HCI	-	Promising
Savelka et al. (2023)	Qual coding (deductive)	augment	NLP	-	Mixed
Zhang et al. (2025)	Qual coding (deductive)	augment	Qual-methods	0.46	Mixed
authors (2025)	Qual coding (deductive)	augment	Qual-methods	0.46	Mixed
Gao et al. (2024)	Qual coding (inductive)	augment	HCI	-	Promising
Dunivin (2024)	Qual coding (inductive)	replace	Qual-methods	-	Promising
Ngo et al. (2026)	Qual coding (inductive)	augment	HCI	-	Mixed
Wu et al. (2023)	Social-computing annotation	replace	CSCW	-	Mixed
Zhu et al. (2023)	Social-computing annotation	replace	CSCW	-	Mixed
Rajashekar et al. (2024)	Social-computing annotation	augment	HCI	-	Mixed
Argyle et al. (2023)	Synthetic participants	simulate	CSS	-	Mixed
Bisbee et al. (2023)	Synthetic participants	replace	CSS	-	Caution
Cheng et al. (2023)	Synthetic participants	simulate	NLP	-	Caution
Wang et al. (2024)	Synthetic participants	replace	HCI	-	Caution
Bail (2024)	Synthetic participants	simulate	CSS	-	Mixed
Abbasiantaeb et al. (2024)	Synthetic participants	simulate	IR	-	Mixed
Huang et al. (2025)	Synthetic participants	simulate	CSS	-	Mixed
Salminen et al. (2025)	Synthetic participants	replace	HCI	-	Caution
De Paoli (2023)	Thematic analysis	augment	CSS	-	Mixed
Dai et al. (2023)	Thematic analysis	augment	NLP	-	Promising
Prescott et al. (2024)	Thematic analysis	augment	Health	-	Mixed
Nguyen-Trung (2025)	Thematic analysis	augment	Qual-methods	-	Mixed
Jain et al. (2025)	Thematic analysis	replace	Qual-methods	0.91	Promising
Castellanos et al. (2025)	Thematic analysis	augment	Health	0.80	Promising
Sharma et al. (2025)	Thematic analysis	augment	HCI	-	Promising
Wen et al. (2025)	Thematic analysis	augment	Qual-methods	-	Mixed
Kocaballi (2023)	UX/design critique	augment	HCI	-	Mixed
Duan et al. (2024)	UX/design critique	augment	HCI	-	Mixed
Duan et al. (2024)	UX/design critique	augment	HCI	-	Mixed
Duan et al. (2024)	UX/design critique	augment	HCI	-	Mixed
Guerino et al. (2025)	Usability/heuristic eval	replace	HCI	0.21	Caution
Zhong et al. (2025)	Usability/heuristic eval	replace	HCI	0.75	Mixed